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**Siyuan Qingneng Electrical and Electronic
Co., LTD. Technical documents**

Siyuan Qingneng High voltage SVG remote
communication
Protocol and Point Table (V1.0)

Siyuan Qingneng Electric & Electronics Co., LTD

Version History

Editions	Author	Participants	Start and end dates	Modifications
V1.0	Wu Bo	Zhang Jian	20231016	Complete the first release of the communication Protocol and the point table document by combining the previous documents and the latest point table

Contents

1. OVERVIEW.....	4
2. TYPE OF STATUTE.....	4
2.1 MODBUS PROTOCOL.....	4
2.2 104 PROTOCOL.....	4
3. MODBUS PROTOCOL TYPICAL PACKET.....	5
3.1 REMOTE MEASUREMENT.....	5
3.2 REMOTE CONFIDENCE.....	6
3.3 DELIVER REMOTE CONTROL.....	7
3.4 PARAMETER SETTING.....	8
4. TYPICAL IEC-104 PROTOCOL MESSAGE.....	11
4.1 TOTAL CALL PROCEDURE.....	11
4.2 TELECOMMAND PROCESS	13
4.3 REMOTE REGULATION PROCESS MESSAGE REMOTE REGULATION PROCESS MESSAGE.....	16
4.4 CHANGE TELEMETERING UPLOAD.....	16
4.5 TRANSPOSITION TELESIGNAL UPLOAD BY TELEMAL.....	17

Communication Protocol

1. Overview

This document specifies the communication protocol between the SVG device and the remote monitoring background, and specifies the mode, format, and content of the communication.

The communication protocol used in this document is a subset of the standard Modbus protocol and a subset of the standard Network 104 protocol. When used, the remote monitoring background is the host and the SVG device is the slave.

Siyuan Qingneng reserves the right to modify the protocol, and will notify the partner manufacturer as soon as possible if there is any modification.

Siyuan Qingneng Electric & Electronics Co., Ltd. reserves the right of final interpretation of the stipulation.

2. Type of Statute

2.1 Modbus Protocol

Physical connection: RS-485;

Transfer rate: Baud rate 9600 bps;

Transmission specifications: 8-bit data bit, 1-bit start bit, 1-bit stop bit, no parity bit;

Communication protocol: A subset of the standard Modbus protocol.

Packet check: Using 16-bit cyclic redundancy check CRC-16;

Other restrictions: **the minimum interval between sending packets is 300ms.**

The Modbus subset in this document uses the following three function codes:

03h: Read hold register (remote measurement);

04h: read input register (remote measurement);

05h: Set a single relay (remote control amount);

06h: set a single hold relay (remote control: single parameter setting);

10h: set multiple hold relays (remote adjustment: multiple parameter Settings);

20h: read hold relays (remote adjustment parameter read back).

2.2 104 Protocol

Physical connection: Network cable;

Communication protocol: Uses a subset of the standard network 104 protocol.

3. Modbus Protocol Typical packet

3.1 Remote Measurement

Modbus function code: 03H

When Modbus reads the analog, the host sends the following packet:

1	Address	
2	Function code	03h
3	The data address is 8 bits higher	
4	The data address is 8 digits lower	
5	The data number is 8 digits higher	00h
6	The data number is 8 digits lower	n
7	The CRC check has the lowest 8 bits	
8	CRC check high 8 bits	

Slave replies are as follows:

1	Address	
2	Function code	03h
3	Bytes of data	2*n
4	Data 1 is 8 digits high	
5	Data 1 Lower 8 digits	
	...	
	Data n is 8 digits higher	
	Data n is 8 digits lower	

	CRC check is 8 bits lower	
	CRC checks the high 8 bits	

3.2 Remote confidence

Modbus function code: 04H

When Modbus reads the status, the host sends a packet as follows:

1	Address	
2	Function code	04h
3	The data address is 8 bits higher	00
4	The data address is 8 digits lower	00
5	The data number is 8 digits higher	00h
6	The data number is 8 digits lower	n
7	CRC check low 8 bits	
8	CRC check high 8 bits	

Slave replies are as follows:

1	Address	
2	Function code	04h
3	Bytes of data	2*n
4	Data 1 is 8 digits high	
5	Data 1 Lower 8 digits	
	...	

	CRC check low 8 bits	
	CRC checks the high 8 bits	

3.3 Deliver remote control

Modbus function code: 05H

Modbus data base address: 0x0000.

When Modbus sets the remote control, the host sends the following message:

1	Address	
2	Function code	05h
3	Set the address 8 bits higher	
4	Set the address 8 digits lower	
5	Set the content 8 bits high	
6	Set the content 8 digits lower	
7	CRC check low 8 bits	
8	CRC check high 8 bits	

The slave returns the following message:

1	Address	
2	Function code	05h
3	Set the address 8 bits higher	
4	Set the address 8 digits lower	
5	Set the content 8 bits high	
6	Set the content 8 digits lower	
7	CRC check low 8 bits	

8	CRC check high 8 bits	
---	--------------------------	--

Note: In the Settings, send FF00 for start (cast) and send 0000 for stop (cut).

3.4 Parameter setting

Modbus function code: 06H, 10H

Base address of Modbus data segment: 0x0000

104 Base value: 0x8020

Number of data: The communication that has been realized so far is shown in Table 4

Description: Operation mode of the control device, reactive power reference, voltage reference value setting, etc

1. When Modbus single parameter is delivered, the host sends the following message:

1	Address	
2	Function code	06h
3	The parameter address is 8 bits higher	
4	Parameter address lower by 8 bits	
5	The parameter content is 8 bits higher	00h
6	Parameter content is 8 digits lower	n
7	CRC check low 8 bits	
8	CRC check high 8 bits	

Parameter Settings The command returned from the machine is as follows:

1	Address	
2	Function code	06h
3	The parameter address is 8 bits higher	
4	Parameter	

	address lower by 8 bits	
5	The parameter content is 8 bits higher	00h
6	Parameter content is 8 digits lower	n
7	The CRC check has the lowest 8 bits	
8	CRC check high 8 bits	

2. When Modbus is delivered with multiple parameters, the host sends the following packet:

1	Address	
2	Function code	10h
3	The parameter address is 8 bits higher	
4	Parameter address lower by 8 bits	
5	The number of arguments is 8 digits higher	00h
6	The number of arguments is 8 digits lower	n
	Number of bytes in the data area	2*n
	Parameter 1 High 8 bits	
	Parameter 1 is 8 bits lower	
	...	
	Parameter n is 8 bits higher	

	Parameter n is 8 bits lower	
	CRC check the lower 8 bits	
	CRC checks the high 8 bits	

Parameter Settings The command returned from the machine is as follows:

1	Address	
2	Function code	10h
3	The parameter address is 8 bits higher	
4	Parameter address lower by 8 bits	
5	The number of arguments is 8 digits higher	00h
6	The number of arguments is 8 digits lower	n
7	CRC check low 8 bits	
8	CRC check high 8 bits	

Table 4 Parameter setting table (intra-segment relative addresses)

Modbus Address	104 address	Name	Multiples	Units	Remarks
0x0000	0x8020	Reactive power control mode	1		0: constant reactive power; 1: constant voltage; 2: constant power factor; 3、System reactive power;
0x0001	0x8021	Constant reactive power instruction	100	Mvar	

0x0002	0x8022	Voltage Reference	10		
0x0003	0x8023	Constant current instruction	1	A	
0x0004	0x8024	Control point 1 device constant reactive power fixed	100	Mvar	
0x0005	0x8025	Control point 2 Device constant reactive power set	100	Mvar	
0x0006	0x8026	Control point 3 Device constant reactive power set	100	Mvar	
0x0007	0x8027	Control point 4 Device constant reactive power set	100	Mvar	

4. Typical IEC-104 protocol message

4.1 Total Call procedure

(1) The master sends a general call message: (

1) The master sends a general call message:

68 0E	00 00	00 00	64	01	06 00	00 01	00 00 00	14
	Send sequence number	Receive sequence number Receive sequence number	Type identification Type Identification	Variable structure qualifier Variable structure qualifier	Reason for transmission	Public address Public address	Message body address Message body address	Distinguish whether it is a general call or a group call

(2) General call confirmation: (

2) General call confirmation:

68 0E	00 00	00 00	64	01	07 00	00 01	00 00 00	14
	Send sequence number Send sequence number	Receive sequence number Receive sequence number	Type identification Type Identification	Variable structure qualifier Variable structure qualifier	Reason for transmission	Public address Public address	Message body address Message body address	Distinguish whether it is a general call or a group call

(3) Send telesignal value: (

3) Send telesignal value:

68

35: ASDU length

								join B1B0: 0,3-not executed,1-open,2-clo se
--	--	--	--	--	--	--	--	--

(2) Telecommand selection confirmation message

. (2) Telecommand selection confirmation message

68 0E	00 00	00 00	2E	01	07 00	00 01	10 60 00	82	
								B7	B6~B2 B1B0
	Send sequenc e number	Receive sequenc e number	Type identification Type	Variable structure qualifier . Variable structure qualifier	Reason for transmissio n	Public address Public address	Telecomman d point: Telecomman d point: SVG start machine SVG start	Telecommand command qualifier: Telecommand command qualifier: B7:1 -execute, 0-select B7:1 -execute, 0-select B1B0: 0,3- Do not execute,1- score,2- join B1B0: 0,3-not executed,1-open,2-close	

(3) Telecommand execution command

. (3) Telecommand execution command

68 0E	00 00	00 00	2E	01	06 00	00 01	10 60 00	02	
								B7	B6~B2 B1B0
	Send sequenc e number	Receive sequenc e number	Type identification Type	Variable structure qualifier . Variable structure qualifier	Reason for transmissio n	Public address Public address	Telecomman d point: Telecomman d point: SVG start machine SVG start	Telecommand command qualifier: Telecommand command qualifier: B7:1 -execute, 0-select B7:1 -execute, 0-select B1B0: 0,3- Do not execute,1- score, 2- join B1B0: 0,3-not executed,1-open, 2-close	

4) Telecommand execution confirmation

. 4) Telecommand execution confirmation

68 0E	00 00	00 00	2E	01	07 00	00 01	10 60 00	02	
								B7	B6~B2 B1B0
	Send sequenc	Receive sequenc	Type identification	Variable structure	Reason for	Public address	Telecomman d point:	Telecommand command qualifier: Telecommand command qualifier:	

	Send sequence number	Receive sequence number	Type Identification	Variable structure qualifier	Reason for transmission	Public address	Telecommand point: SVG start machine SVG start	B7:1 -execute, 0-select B7:1 -execute, 0-select B1B0: 0,3- Do not execute,1- score,2-join B1B0: 0,3-not executed,1-open,2-close
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(5) Telecommand cancellation command

. (5) Telecommand cancellation command

68 0E	00 00	00 00	2E	01	08 00	01 00	10 60 00	02	
								B7	B6~B2 B1B0
	Send sequence number	Receive sequence number	Type Identification	Variable structure qualifier	Reason for transmission	Public address	Telecommand point: SVG start machine SVG start	Telecommand command qualifier: Telecommand command qualifier: B7:1 -execute, 0-select B7:1 -execute, 0-select B1B0: 0,3- Do not execute,1- score,2-join B1B0: 0,3-not executed,1-open,2-close	

(6) Remote undo confirmation

(6) Telecommand cancel confirmation

68 0E	00 00	00 00	2E	01	09 00	00 01	10 60 00	02	
								B7	B6~B2 B1B0
	Send sequence number	Receive sequence number	Type Identification	Variable structure qualifier	Reason for transmission	Public address	Telecommand point: SVG start machine SVG start	Telecommand command qualifier: Telecommand command qualifier: B7:1 -execute,0-select B7:1 -execute,0-select B1B0:0,3- Do not execute,1- score,2-join B1B0:0,3-not executed,1-open,2-close	

4.3 Remote regulation process message

Remote regulation Process Message

(1) Host remote regulation message

(1) Master Remote Regulation Message

68 0E	00 00	00 00	31	01	06 00	00 01	01 (di) 62(gao) 00	00(di) 02(gao)	00
	Send sequence number. Send sequence number	Receive sequence number Receive sequence number	Type identification: Type identification: Scaled set value. Scaled Set Value	Variable structure qualifier . Variable structure qualifier	Reason for transmissio n	Public address Public address	Message body address Message body address	Set value Set value	

Remote regulation confirmation message

(2) Remote regulation confirmation message

68 0E	00 00	00 00	31	01	07 00	00 01	01 62 00	00 02	00
	Send sequenc e number. Send sequenc e number	Receive sequence number Receive sequence number	Type identification: Type identification: Scaled set value. Scaled Set Value	Variable structure qualifier . Variable structure qualifier	Reason for transmissio n	Public addres s Public addres s	Message body address Message body address	Set value Set value	

4.4 Change telemetering upload

68

16: ASDU length

16: ASDU length

0C 00: Send sequence number.

0C 00: Send sequence number

04 00: Receive sequence number.

04 00: Receive sequence number

09: Type identification-telemetering with quality description

09: Type identification-telemetering with quality description

02: Variable structure qualifier, SQ=0 -- addressed by information object address; 02: Variable structure qualifier, SQ=0-addressed

by the address of the information object; 02: variable structure qualifier, sq =0-addressed by the address of the information object; 2 changes telemetering

03 00: reason for transmission

. 03 00: reason for transmission

01 00: public address

01 00: public address

01 40 00: message body address -- 0x400101

40 00: message body address -- 0x4001

12 34 00: First remote measurement, 12 34-- telemetry value (0x3412), 00 -- Quality Factor Description

12 34 00: the first telemetering data 12 34-telemetry value (0x3412), 00-quality factor description

05 40 00: message body address -- 0x400505

40 00: message body address -- 0x4005

78 56 00: Second remote measurement, 78 56-- Telemetry value (0x5678), 00 -- Quality Factor Description

78 56 00: the second telemetering data, 78 56-telemetry value (0x5678), 00-quality factor description

4.5 Transposition telesignal upload by telemail

68

12: ASDU length

12: ASDU length

0C 00: Send sequence number.

0C 00: Send sequence number

04 00: Receive sequence number.

04 00: Receive sequence number

01: Type identification-single point telesignal

. 01: Type identification-single point telesignal

02: Variable structure qualifier, SQ=0 -- addressable by information object address; 02: Variable structure qualifier, SQ=0-addressed

by the address of the information object; 02: variable structure qualifier, sq =0-addressed by the address of the information object; 2 changes telemetering

03 00: reason for transmission

. 03 00: reason for transmission

01 00: public address

01 00: public address

01 00 00: message body address -- 0x000101

00 00: message body address -- 0x0001

00: The first telesignal data, open-the main circuit breaker

is open. 00: The first telesignal data, open-the main circuit breaker is open

05 00 00: message body address -- 0x000505

00 00: message body address -- 0x0005

01: The second elesignal data, close-the device is locked

._01: The second elesignal data, close-the device is locked